

CARDIORESPIRATORY AND CARDIAC AUTONOMIC RESPONSES TO 30–15 INTERMITTENT FITNESS TEST IN TEAM SPORT PLAYERS

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ABSTRACT

Buchheit, M, Al Haddad, H, Millet GP, Lepretre, PM, Newton, M, and Ahmaidi, S. Cardiorespiratory and cardiac autonomic responses to 30–15 Intermittent Fitness Test in team sport players. *J Strength Cond Res* 23(1): 93–100, 2009—The 30–15 Intermittent Fitness Test (30–15_{IFT}) is an attractive alternative to classic continuous incremental field tests for defining a reference velocity for interval training prescription in team sport athletes. The aim of the present study was to compare cardiorespiratory and autonomic responses to 30–15_{IFT} with those observed during a standard continuous test (CT). In 20 team sport players (20.9 ± 2.2 years), cardiopulmonary parameters were measured during exercise and for 10 minutes after both tests. Final running velocity, peak lactate ([La]_{peak}), and rating of perceived exertion (RPE) were also measured. Parasympathetic function was assessed during the postexercise recovery phase via heart rate (HR) recovery time constant (HRR_τ) and HR variability (HRV) vagal-related indices. At exhaustion, no difference was observed in peak oxygen uptake ($\dot{V}O_{2peak}$), respiratory exchange ratio, HR, or RPE between 30–15_{IFT} and CT. In contrast, 30–15_{IFT} led to significantly higher minute ventilation, [La]_{peak}, and final velocity than CT ($p < 0.05$ for all parameters). All maximal cardiorespiratory variables observed during both tests were moderately to well correlated (e.g., $r = 0.76$, $p = 0.001$ for $\dot{V}O_{2peak}$). Regarding ventilatory thresholds (VThs), all cardiorespiratory measurements were similar and well correlated between the 2 tests. Parasympathetic function was lower after 30–15_{IFT} than after CT, as indicated by significantly longer

HRR_τ (81.9 ± 18.2 vs. 60.5 ± 19.5 for 30–15_{IFT} and CT, respectively, $p < 0.001$) and lower HRV vagal-related indices (i.e., the root mean square of successive R-R intervals differences [rMSSD]: 4.1 ± 2.4 and 7.0 ± 4.9 milliseconds, $p < 0.05$). In conclusion, the 30–15_{IFT} is accurate for assessing VThs and $\dot{V}O_{2peak}$, but it alters postexercise parasympathetic function more than a continuous incremental protocol.

KEY WORDS aerobic power, field test, heart rate variability, intermittent exercise, postexercise parasympathetic reactivation

INTRODUCTION

High-intensity aerobic training performed in team sports often includes intermittent and shuttle runs to reproduce running patterns specific to competition (15,19). The 30–15 Intermittent Fitness Test (30–15_{IFT}) (9), a graded intermittent and shuttle field test, leads to a maximal running velocity (V_{IFT}) that is determined through an effort involving physiological variables similar to those solicited during shuttle interval training sessions (i.e., ability to change direction and both aerobic and anaerobic energy systems). The use of V_{IFT} has been shown to be more accurate than individual $\dot{V}O_{2max}$ (defined as the lowest running velocity that elicits maximal oxygen uptake during a continuous graded test [5]) for getting players with different physiological profiles to a similar level of cardiorespiratory demand and, thus, for standardizing training content at a team level (9). Nevertheless, the physiological responses induced by the 30–15_{IFT} have not been compared with a gold-standard continuous incremental test (CT) (2,23) for identifying submaximal (i.e., ventilatory thresholds) and maximal cardiorespiratory variables. To date, it has only been shown that V_{IFT} is very well related to maximal oxygen uptake determined in the laboratory and that maximal heart rate (HR) (9) is achieved.

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Assessment of postexercise autonomic regulation has recently emerged as an original way to estimate the short-term physiological strain of exercise, because it can indirectly provide insight into perturbation of body regulatory processes and homeostasis (10,11). Postexercise autonomic assessment can also transiently highlight increased cardiovascular risk (6,7,13) in specific populations (i.e., sedentary individuals engaging in rehabilitation programs). Postexercise parasympathetic activity has never been described after any graded intermittent exercise. Nevertheless, because supra-maximal and intermittent exercise has been shown to strongly impair postexercise autonomic function (10) and the body's homeostasis (20), it could be expected that the 30–15_{IFT}, leading to higher running velocities than $\dot{V}O_{2\max}$ (9), would have a more depressive effect on postexercise autonomic function than CT.

The aim of the present study was thus to compare cardiorespiratory and autonomic responses between the 30–15_{IFT} and a graded continuous field test.

METHODS

Experimental Approach to the Problem

Twenty moderately trained healthy men's team sport players were recruited (20.9 ± 2.2 years, 179.8 ± 5.8 cm, and 76.4 ± 9.4 kg, playing soccer, handball, or basketball and training 4.9 ± 1.9 h·wk⁻¹). They all provided informed consent to participate in the study, which conformed to the recommendations of the Declaration of Helsinki and was approved by the local institutional research ethics committee.

Subjects

All subjects were familiar with field exercise evaluation and were tested on 2 separate days on a synthetic indoor track (temperature ≈ 22 °C). The 30–15_{IFT} consists of 30-second shuttle runs interspersed with 15-second passive recovery periods. Velocity was set at 8 km·h⁻¹ for the first 30-second run, and speed was increased by 0.5 km·h⁻¹ every 30-second stage thereafter (9). Subjects were required to run back and forth between 2 lines set 40 m apart at a pace that was governed by a prerecorded beep. During the 15-second recovery period, subjects walked in a forward direction toward the closest line (at either the middle or end of the running area, depending on where their previous run had stopped) and commenced the next run stage from this line. The test has been shown to be highly reliable (intraclass correlation coefficient [ICC] = 0.96 for the final running speed) (9). The graded continuous field test was adapted from the extensively used test of Léger and Boucher (23): the initial speed of 10 km·h⁻¹ increased by 0.5 km·h⁻¹ per 1-minute stage, with subjects running at a pace that was governed by a prerecorded beep. The test has been shown to be highly reliable ($r = 0.97$) (23). The velocity (km·h⁻¹) attained during the last completed stages of each test was determined as the subject's V_{IFT} and V_{CT} for 30–15_{IFT} and CT, respectively.

Physiological Measurements

Respiratory gas exchange and HR were measured using an automated breath-by-breath system (K4b², Cosmed, Rome, Italy), which was calibrated as recommended by the manufacturer before each test (14). Cardiorespiratory data—oxygen uptake ($\dot{V}O_2$), carbon dioxide production ($\dot{V}CO_2$), breathing frequency (BF), tidal volume (VT), minute ventilation ($\dot{V}E$), respiratory exchange ratio (RER), O_2 pulse, end tidal volume PO_2 (PET_{O2} %), PCO_2 (PET_{CO2} %), and HR—were filtered and averaged on a 5-second basis. For both tests, peak oxygen uptake ($\dot{V}O_{2peak}$) was arbitrarily defined as the highest $\dot{V}O_2$ value attained in a 20-second period (obtained from 4 averaged 5-second data). Subjects' $\dot{V}CO_2$, BF, VT, $\dot{V}E$, and RER at $\dot{V}O_{2peak}$ were retained as $\dot{V}CO_{2peak}$, BF_{peak}, VT_{peak}, $\dot{V}E_{peak}$, RER_{peak}, and O_2 pulse peak. An HR_{peak} attained coincident with the maximal predicted value, peak blood lactate ($[La]_{bpeak}$) > 8 mmol·L⁻¹, RER > 1.1, and volitional fatigue were required to confirm the maximal nature of the test. The velocity associated with $\dot{V}O_{2peak}$ ($\dot{V}O_{2peak30-15IFT}$ and $\dot{V}O_{2peakCT}$) were the lowest running velocities that elicited a $\dot{V}O_2$ value equal to $\dot{V}O_{2peak}$ during the 30–15_{IFT} and CT, respectively (5). Three minutes after the end of each exercise set, a fingertip blood sample (5 μ l) was collected in the 20 subjects ($[La]_{bpeak}$, Lactate Pro, Arkray Inc, Kyoto, Japan). Additionally, every 4 stages during the 30–15_{IFT} (i.e., every 2 km·h⁻¹ during the 15-second recovery periods), lactate profile was determined in 11 of the 20 subjects. The accuracy of the analyzer was checked before each test using standards.

Ventilatory Thresholds Assessment

Cardiorespiratory data were averaged at each stage level. For the 30–15_{IFT}, data from the 30-second exercise and the successive 15-second recovery periods were averaged to produce a single 45-second stage value. As described by Whipp et al. (34), the first ventilatory threshold (V_{Th1}) was established as the velocity at which $\dot{V}E/\dot{V}O_2$ and PET_{O2} began to increase without a simultaneous increase in PET_{CO2}. The second ventilatory threshold (V_{Th2} [also called RCP, respiratory compensation point]) was determined as the velocity corresponding to the first decrease in PET_{CO2}, with a corresponding increase in $\dot{V}E/\dot{V}CO_2$ after the steady-state phase after the V_{Th1} . All threshold measurements were made by visual inspection of graphs of time plotted against each relevant respiratory variable measured in the testing. The visual inspections were made by 2 experienced exercise physiologists, and the results were compared and then averaged. The difference in the values determined by the 2 assessors was <2%.

Beat-to-Beat Heart Rate Recordings

Beat-to-beat HR data were additionally recorded with a Polar S810 HR monitor (Polar Electro, Kempele, Finland) (16). All R-R interval series were extracted (Polar Protrainer 5.1, Polar Electro), and occasional ectopic beats were visually identified

and manually replaced with interpolated adjacent R-R interval values. As previously described (11), parasympathetic function was assessed during the 10-minute period after both CT and 30–15_{IFT} by 1) analyzing the first 30 seconds (from the 10th to the 40th seconds) of HR recovery (HRR) via semilogarithmic regression analysis (T30, short-term time constant), 2) by taking the complete time constant of the HR decay obtained by fitting the 10-minute postexercise HR recovery into a first-order exponential decay curve (HRR_τ), 3) by calculating a time-varying vagal-related index (the root-mean-square difference of successive normal R-R intervals on subsequent 30-second nonoverlapped segments, rMSSD_{30s}), or 4) by analyzing usual stationary vagal-related HRV indices (11), calculated as the last 5 minutes of the 10-minute recovery period in the seated position. Mean HR (HR_{5–10 min}), SD of normal R-R intervals (SDNN_{5–10 min}), rMSSD_{5–10 min}, power densities in the low (LF, 0.04–0.15 Hz) and high-frequency band (HF, >0.15–0.50 Hz), LnHF_{5–10 min} (natural log of HF), and normalized LF power (LFnu_{5–10 min}, calculated as the LF / (LF + HF) ratios) were computed (33). All indices were considered to provide insight into parasympathetic activity, except LFnu_{5–10 min}, which was used as a relative index of the sympathetic component of sympathovagal balance (33).

Statistical Analyses

Descriptive data are presented as mean ± SD. Normality of the distribution was verified via the Shapiro Wilk test. Student's *t*-test for paired data, Pearson coefficient correlation (*r*) calculated from linear regressions, and Bland-Altman limits of agreement (8) were used to compare ventilatory measures, [La]_bpeak, and RPE measures between 30–15_{IFT}

and CT at the 3 intensities (V_{Th1}, V_{Th2}, and $\dot{V}O_{2peak}$). Bland-Altman plots were presented to illustrate the intra-individual agreement in $\dot{V}O_{2peak}$ and HRpeak for the 30–15_{IFT} and CT (8). The HR(V)-derived indices and running velocity were compared using a paired Student's *t*-test and Pearson coefficient correlation. For time-varying rMSSD_{30s}, a 2-factor repeated-measures analysis of variance (ANOVA) with one between factor (test type; 30–15_{IFT} vs. CT) and a within factor (time; 19 repeated measures during the recovery) was used to examine the main effects and/or interactions of test type and time during recovery. For each ANOVA, if a significant interaction was identified, a Bonferroni post hoc test was used to further delineate differences between training type, period, and time. The level of significance was set at *p* ≤ 0.05. Statistical analyses were carried out using Minitab 14 Software (Minitab Inc, Paris, France).

RESULTS

Velocities at exhaustion, and first and second ventilatory thresholds, were all higher during 30–15_{IFT} compared with CT: 19.4 ± 0.9 vs. 15.4 ± 1.1 km·h⁻¹ (*p* < 0.001) for V_{IFT} and V_{CT}, respectively; 18.3 ± 1.3 vs. 14.1 ± 1.4 km·h⁻¹ (*p* < 0.001) for V_{IFT} $\dot{V}O_{2peak}$ and V_{CT} $\dot{V}O_{2peak}$, respectively; 16.9 ± 1.3 vs. 13.6 ± 1.3 km·h⁻¹ (*p* < 0.001) for 30–15_{IFT} and CT at V_{Th2}, respectively; and 13.1 ± 1.2 vs. 10.1 ± 0.6 km·h⁻¹ (*p* < 0.001) for 30–15_{IFT} and CT at V_{Th1}, respectively.

Mean maximal cardiorespiratory responses of both tests are presented in Table 1. Figure 1 illustrates the Bland-Altman plot for $\dot{V}O_{2peak}$ and HRpeak. All values remained in the 95% confidence interval. Values for $\dot{V}O_{2peak}$, $\dot{V}CO_{2peak}$,

TABLE 1. Maximal values of cardiovascular variables of interest for the 30–15 Intermittent Fitness Test (30–15_{IFT}) and the continuous graded test (CT).

	30–15 _{IFT}	CT	95% CI for the mean difference	<i>r</i> (<i>p</i>)
$\dot{V}O_{2peak}$ (mL·min ⁻¹ ·kg ⁻¹)	54.5 ± 6.6	53.5 ± 7.3	1.60; 3.01	0.76 (0.001)
BFpeak (cycles per minute)	54.0 ± 10.8*	48.1 ± 8.9	–0.85; 10.96	0.53 (0.04)
VTpeak (L)	2.7 ± 0.3	2.7 ± 0.4	0.23; 0.14	0.58 (0.01)
$\dot{V}E_{peak}$ (L·min ⁻¹)	141.4 ± 20.1*	128.7 ± 22.1	–3.26; 22.19	0.60 (0.01)
$\dot{V}CO_{2peak}$ (mL·min ⁻¹ ·kg ⁻¹)	60.3 ± 9.7*	56.4 ± 6.4	–0.57; 7.26	0.77 (0.001)
RERpeak	1.11 ± 1.1	1.06 ± 0.1	0.02; 0.10	0.56 (0.05)
O ₂ pulse peak (mL per beat)	22.1 ± 2.9	21.3 ± 3.1	0.44; 2.09	0.64 (0.004)
HRpeak (bpm)	195.0 ± 8.4	192.5 ± 10.1	0.42; 5.19	0.84 (0.000)
[La] _b peak (mmol·L ⁻¹)	11.8 ± 2.3†	8.9 ± 2.1	–1.44; 4.28	0.14 (0.54)
RPE	7.1 ± 1.1	6.6 ± 1.2	0.30; 1.21	0.55 (0.08)

Values are mean (± SD). Confidence interval (CI) for the mean difference (30–15_{IFT} minus CT) and Pearson coefficient correlation (*r*) are also presented.

$\dot{V}O_{2peak}$ = peak oxygen uptake; BFpeak = peak breathing frequency; VTpeak = peak tidal volume; $\dot{V}E_{peak}$ = peak minute ventilation; $\dot{V}CO_{2peak}$ = peak carbon dioxide production; RERpeak = peak respiratory exchange ratio; O₂pulse peak = oxygen pulse at $\dot{V}O_{2peak}$; HRpeak = peak heart rate; [La]_bpeak = peak blood lactate concentration; RPE = rating of perceived exertion.

*Significant difference compared with CT (*p* < 0.05); †significant difference compared with CT (*p* < 0.01).

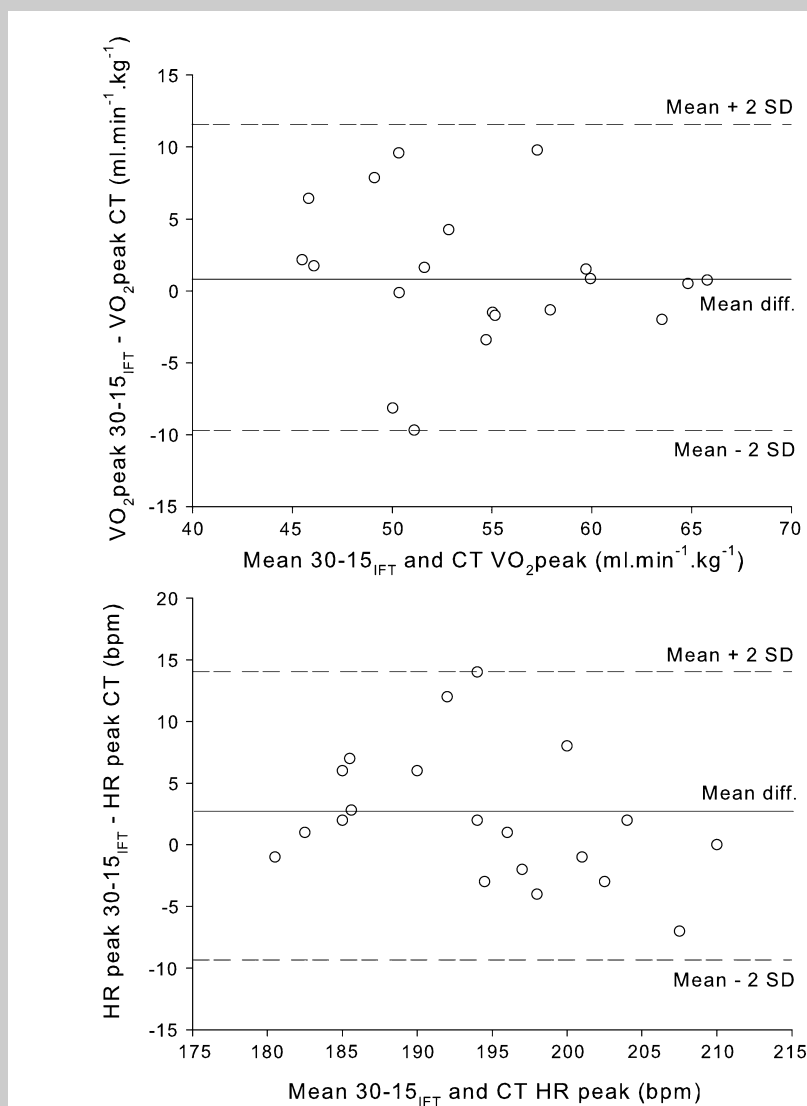


Figure 1. Bland-Altman plot of peak oxygen uptake ($\dot{V}O_{2peak}$) and peak heart rate (HR_{peak}) measured for the 30-15_{IFT} vs. continuous test (CT).

$\dot{V}E_{peak}$, and $[La]_{bpeak}$ measured for 30-15_{IFT} as a percentage of CT are shown in Figure 2. It seems that whereas $\dot{V}O_{2peak}$ was similar, $\dot{V}CO_{2peak}$, $\dot{V}E_{peak}$, and $[La]_{bpeak}$ were significantly higher in 30-15_{IFT} compared with CT (all $p < 0.05$). There were no differences in absolute values for $\dot{V}O_{2peak}$, VT_{peak} , RER_{peak} , O_2 pulse peak, HR_{peak}, and RPE between 30-15_{IFT} and CT. In contrast, 30-15_{IFT} led to significantly higher BF_{peak}, $\dot{V}E_{peak}$, and $[La]_{bpeak}$ than CT. Pearson correlation coefficients show (Table 1) that all maximal cardiorespiratory variables observed during both tests were moderately to well correlated (e.g., $r = 0.76$ and $p = 0.001$ for $\dot{V}O_{2peak}$). For the 11 subjects in whom blood samples were collected during the 30-15_{IFT}, mean blood lactate concentration was 1.1 ± 0.1 mmol·L⁻¹ at rest, and it

increased with running velocity to reach 12.3 ± 2.8 mmol·L⁻¹ at 3 minutes after the end of exercise (Figure 3).

With regard to ventilatory thresholds, all cardiorespiratory measurements were similar between the 2 tests (Table 2). Pearson correlation coefficients show (Table 2) that all cardiorespiratory variables at both ventilatory thresholds were moderately to well correlated.

Parasympathetic function was appreciably more impaired after 30-15_{IFT} than after CT, as indicated by significantly longer T30 ($p < 0.05$), longer HHR τ ($p < 0.001$), lower rMSSD_{5-10 min} ($p < 0.01$), and decreased LnHF_{5-10 min} ($p < 0.01$) (Table 3). Conversely, LFnu_{5-10 min} values indicated a higher shift toward sympathetic predominance after 30-15_{IFT} than CT ($p < 0.05$). The HHR τ and rMSSD_{5-10 min} observed after the 30-15_{IFT}, expressed as percentages of the values observed after CT, are presented in Figure 2. The rMSSD_{30s} did not display any significant change during the 10-minute recovery period after 30-15_{IFT}, whereas it increased significantly after CT (Figure 4).

DISCUSSION

The present study is the first to describe cardiorespiratory and autonomic responses to a shuttle and intermittent field test, the 30-15_{IFT}. Peak oxygen uptake and most cardiorespiratory parameters at the 2 ventilatory thresholds were similar for the 30-15_{IFT} and classic graded continuous field test. In contrast, the 30-15_{IFT} elicited higher peak minute ventilation and maximum carbon dioxide production, which could be related to the final higher running velocity reached and the increased anaerobic contribution in 30-15_{IFT} compared with CT. Finally, postexercise autonomic function was characterized by a decreased parasympathetic activity and a greater sympathetic predominance after the 30-15_{IFT} than after CT.

Continuous and linear exercise testing is not the optimal method for determining a reference velocity for scheduling shuttle interval training sessions in team sport players (9).

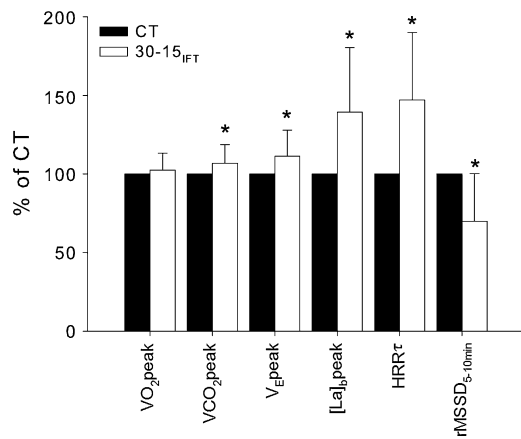


Figure 2. Mean ($\pm$ SD) value of peak oxygen uptake ($\dot{V}O_{2\text{peak}}$), carbon dioxide uptake ($\dot{V}CO_{2\text{peak}}$), and minute ventilation ($\dot{V}E_{\text{peak}}$) at $\dot{V}O_{2\text{peak}}$, peak blood lactate concentration ($[La]_b$), long-term heart rate recovery time constant (HRR τ), and root mean square of successive R-R interval differences (rMSSD_{5-10min}) for 30-15IFT (white bars), expressed as percentages of values observed for CT (black bars). *Significant difference vs. CT ($p < 0.05$).

Therefore, there is a need for designing protocols such as the 30-15_{IFT} that reproduce the characteristics of specific intermittent exercises practiced in those sports (i.e., change of direction, acceleration, deceleration, and rest periods).

Our results demonstrate the validity of the 30-15_{IFT} for assessing maximal aerobic power. We observed that the 30-15_{IFT} elicits similar peak O₂ uptake, VT, RER, O₂ pulse, and HR to a classic graded continuous field test, which we used as a standard reference of aerobic power assessment (Table 1) (2,27). The parameters obtained for the 2 tests were all

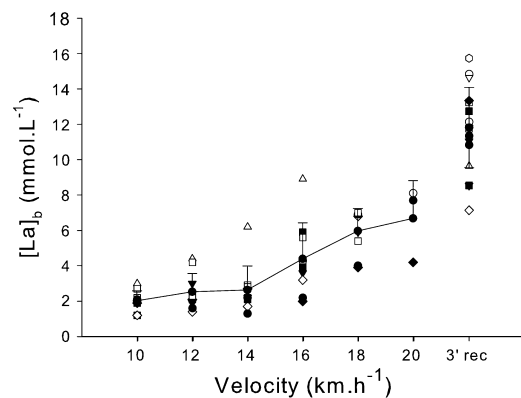


Figure 3. Evolution of blood lactate concentration ($[La]_b$) during the 30-15_{IFT} and 3 minutes after test termination. Mean $\pm$ SD and individual values for 11 subjects are reported.

TABLE 2. Cardiovascular variables of interest at the first (V_{Th1}) and second ventilatory threshold (V_{Th2}) for the 30-15 Intermittent Fitness Test (30-15_{IFT}) and the continuous graded test (CT).

	V_{Th1}				V_{Th2}			
	30-15 _{IFT}	CT	95% CI for the mean difference	r (p)	30-15 _{IFT}	CT	95% CI for the mean difference	r (p)
$\dot{V}O_2$ (mL·min ⁻¹ ·kg ⁻¹)	38.6 $\pm$ 4.7	39.8 $\pm$ 4.8	3.48; 0.95	0.67 (0.01)	49.8 $\pm$ 6.6	50.3 $\pm$ 6.9	3.00; 2.00	0.72 (0.001)
BF (cycles per minute)	32.8 $\pm$ 5.2	33.6 $\pm$ 6.1	3.84; 0.31	0.73 (0.001)	44.2 $\pm$ 6.9	43.0 $\pm$ 6.3	2.01; 4.39	0.54 (0.02)
VT (L)	2.2 $\pm$ 0.4	2.2 $\pm$ 0.5	0.14; 0.13	0.84 (0.000)	2.6 $\pm$ 0.3	2.7 $\pm$ 0.4	0.22; 0.07	0.71 (0.002)
$\dot{V}E$ (L·min ⁻¹)	70.3 $\pm$ 11.6	73.44 $\pm$ 9.4	7.86; 1.65	0.60 (0.03)	114.2 $\pm$ 13.2	113.4 $\pm$ 14.4	7.42; 9.05	0.76 (0.01)
$\dot{V}CO_2$ (mL·min ⁻¹ ·kg ⁻¹)	35.47 $\pm$ 5.3	37.2 $\pm$ 4.1	4.35; 0.88	0.56 (0.04)	52.5 $\pm$ 9.7	53.1 $\pm$ 6.6	3.84; 2.53	0.80 (0.000)
RER	0.91 $\pm$ 0.1	0.93 $\pm$ 0.1	0.05; 0.01	0.66 (0.004)	1.05 $\pm$ 0.1	1.06 $\pm$ 0.1	0.07; 0.05	0.58 (0.01)
O ₂ pulse (mL per beat)	18.5 $\pm$ 2.5	18.5 $\pm$ 3.2	1.52; 1.44	0.53 (0.05)	20.8 $\pm$ 3.0	20.5 $\pm$ 3.3	0.97; 1.46	0.69 (0.001)
HR (bpm)	159.6 $\pm$ 13.5	161.0 $\pm$ 14.4	4.82; 1.92	0.82 (0.000)	182.9 $\pm$ 9.7	183.6 $\pm$ 10.3	3.68; 2.30	0.79 (0.000)

Values are mean ($\pm$ SD). Confidence interval (CI) for the mean difference (30-15_{IFT} minus CT) and Pearson coefficient correlation (r) are also presented. $\dot{V}O_2$ = oxygen uptake; BF = breathing frequency; VT = tidal volume; $\dot{V}E$ = minute ventilation; $\dot{V}CO_2$ = carbon dioxide production; RER = respiratory exchange ratio; O₂pulse = oxygen pulse; HR = heart rate.

*Significant difference compared with CT ($p < 0.05$).

TABLE 3. Postexercise parasympathetic indices calculated for the 30-15 Intermittent Fitness Test (30-15_{IFT}) and the continuous graded test (CT).

	30-15 _{IFT}	CT	95% CI for the mean difference
T30 (s)	261.4 ± 65.9*	221.8 ± 65.9	-4.98; 84.19
HRR _T (s)	81.9 ± 18.2‡	60.5 ± 19.5	11.27; 31.70
SDNN _{5-10 min} (ms)	13.8 ± 6.3*	18.6 ± 7.6	-9.45; -0.10
rMSSD _{5-10 min} (ms)	4.1 ± 2.4†	7.0 ± 4.9	-5.04; -0.70
LnHF _{5-10 min} (Ln ms ²)	1.6 ± 1.0*	2.3 ± 1.5	-1.38; -0.05
LFnu _{5-10 min} (Ln ms ²)	0.86 ± 0.1*	0.77 ± 0.1	0.01; 0.21
HR _{5-10 min} (bpm)	110.6 ± 10.2*	105.5 ± 14.1	-0.72; 11.00

Mean (± SD) values calculated for the 5-minute period, and confidence interval (CI) for the mean difference (30-15_{IFT} minus CT).

T30 = short-term HR time constant; HRR_T = long-term time constant of the HR decay; SDNN_{5-10 min} = standard deviation of normal R-R intervals; rMSSD_{5-10 min} = root-mean-square difference of successive normal R-R intervals; LnHF_{5-10 min} = natural-log of high-frequency power density; LFnu = normalized low-frequency power; HR_{5-10 min} = mean HR.

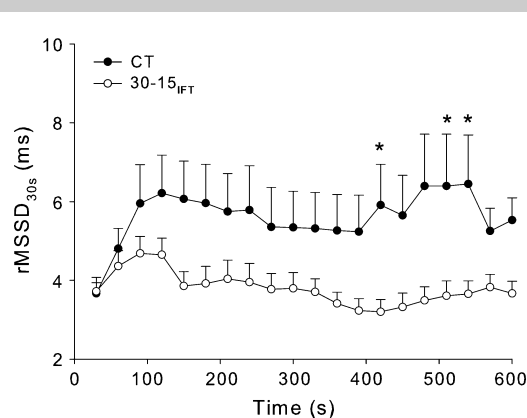
*Significant difference vs. CT ($p < 0.05$); †significant difference vs. CT ($p < 0.01$); ‡significant difference vs. CT ($p < 0.001$).

moderately to well correlated, and 95% intervals of agreement (i.e., Figure 1 for $\dot{V}O_{2peak}$ and HR_{peak}) were similar to those usually reported for comparisons of field tests (12,26). Mean $\dot{V}O_{2peak}$ bias (-1.6; 3.1 mL·min⁻¹·kg⁻¹) was within the range of $\dot{V}O_{2peak}$ interdiary variations (2.4-5.9%) (28) or that of technical error measurement (~3%) (14). The present results can be compared with previous studies of young soccer players (12, 26) that have shown incremental intermittent testing (i.e., the Yo-Yo intermittent endurance test, which employs a protocol similar to the 30-15_{IFT}) to elicit $\dot{V}O_{2peak}$ or HR_{peak} values similar to those observed during standard treadmill CT. Our results contrast, however,

with those reported by Girard et al. (17,18) using specific intermittent tennis and squash tests, because $\dot{V}O_{2peak}$ was higher during intermittent racket tests than those on a treadmill. The higher oxygen consumption observed in the racket sport tests (16,17) may be attributable to the involvement of upper-body muscles required for the simulated ball-hitting action. We, however, observed significantly higher BF_{peak}, $\dot{V}E_{peak}$, and $\dot{V}CO_{2peak}$ during the 30-15_{IFT} than during the CT. One may assume that the discontinuous or shuttle (1) nature of the 30-15_{IFT} partly explains the higher ventilatory parameters observed, because the coupling of locomotion-ventilation is altered by the changes in direction, stride patterns, and velocity. Signals from neural afferents in contracting muscles as a result of increased V_{IFT} compared with V_{CT} may have also led to an excitatory effect on central respiratory drive.

Although blood lactate may not be the best measure to demonstrate the relative contribution of anaerobic energy supply in sports with discontinuous exercise of long duration (32), its significantly higher concentration observed after the 30-15_{IFT} could also account for the increase in ventilation parameters. Metabolic acidosis may have contributed to an enhancement of peripheral chemoreceptor activity resulting in increased ventilation (25). The significantly higher CO_2 production observed for 30-15_{IFT} may also be directly related to higher blood buffering of protons (H^+) compared with CT.

In terms of assessing ventilatory thresholds, the present results show that the 30-15_{IFT} produced values comparable with those of CT. All cardiorespiratory parameters were similar and well correlated at the 2 ventilatory thresholds, and the confidence intervals were acceptable (Table 2). With the exception of the 2 specific racket test studies (17,18) that report similar VThs values for field intermittent and treadmill tests, previous running studies have paid exclusive attention

**Figure 4.** Average (± SD) root mean square of successive differences in the R-R intervals measured on successive 30-second segments (rMSSD_{30s}) during the 10-minute recovery period after the 30-15 Intermittent Fitness Test (30-15_{IFT}) or the graded continuous test (CT).

*Significant difference vs. rMSSD_{30s} at 30 seconds after the end of exercise during the same test session ($p < 0.05$).

to maximal cardiorespiratory values (12,26). Because the intermittent nature of exercise is known to increase anaerobic participation (3,21), the accuracy of the 30–15_{IFT} to predict V_{Ths} was difficult to gauge. However, data suggest that anaerobic participation during intermittent work might only affect ventilatory measures at high intensities (higher than V_{Th2}). Thus, the present results may also encourage the use of the 30–15_{IFT} for V_{Ths} assessments.

This is also the first study to describe HR recovery and postexercise autonomic function after a graded intermittent test. The time-varying vagal-related rMSSD_{30s} did not display any significant increase during the 10-minute recovery period after 30–15_{IFT}, whereas it showed a time-effect increase after CT, indicative of effective parasympathetic reactivation (Figure 4). This apparently reduced parasympathetic modulation (and, conversely, increased sympathetic activity) after the 30–15_{IFT} compared with CT was corroborated by the significantly slower HRR time constants (T30 and HRR_τ), the lower vagal-related HRV indices, and the higher sympathetic-related data (Table 3). Nevertheless, any present inference to a parasympathetic effect on HR should be considered with caution. Various investigations have demonstrated, especially at high workloads, complex interactions between the sympathetic and vagal systems with respect to HR regulation resulting in reduced (30) or amplified vagal stimulation (22,24).

The reduced parasympathetic modulation after 30–15_{IFT} is in agreement with previous data showing that anaerobic participation and muscle power engagement, rather than aerobic intensity per se, influence the postexercise parasympathetic reactivation (10). Although the direct link between HRV and homeostasis has not been objectively determined, we can suggest that the particular autonomic status observed after the 30–15_{IFT} might be related to a more disturbed postexercise homeostasis (i.e., increased system stress metabolite accumulation [31]) as confirmed by the higher blood lactate concentration after the 30–15_{IFT}. This is in line with results reported earlier, where frequent changes in exercise intensity during 40 minutes of cycling were shown to disturb homeostasis and increase postexercise metabolism to a greater extent than constant exercise conducted at a similar mean intensity (21). All vagal HR-derived indices observed here were higher than those reported (10) after 12 minutes of high-intensity exercise (30 seconds of running at 90% of V_{IFT}, interspersed with 30 seconds of passive recovery) in comparable subjects using extremely similar methods (i.e., T30 = 81.9 ± 18.2 and 100.99 ± 7.02 seconds after 30–15_{IFT} and high-intensity exercise) (10). This indicates that the 30–15_{IFT} has an important impact on homeostasis and autonomic nervous system activity but that it is of lower magnitude than traditional interval training exercises (4,29). Because it is believed that a background low in parasympathetic activity after exercise is related to a transiently increased risk for adverse cardiovascular events (6,7,13,33), the present data suggest limiting the use of the 30–15_{IFT} to

team sport athletes and employing the traditional CT in sedentary or unfit subjects for aerobic function assessment. However, as per exercise intensity prescription (4,29), the use of the 30–15_{IFT} could be indicated for the latter stages of a (rehabilitation) training program in sedentary individuals or patients.

PRACTICAL APPLICATIONS

The 30–15_{IFT} protocol seems as accurate as a classic graded continuous test for the assessment of peak oxygen uptake and ventilatory thresholds. These data support the use of the 30–15_{IFT}, not only as an accurate training instrument but also as a valid tool to explore aerobic function. Nevertheless, the slower HR recovery and increased sympathetic predominance after the 30–15_{IFT} suggest that its physiological strain is higher than that of CT. The appreciably depressed post-exercise parasympathetic activity encourages restricting its use to already trained athletes.

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